

Python Data Structures and Program Logic

Represent collections of values and control which instructions execute.



WHY THIS MATTERS

Start from what you already know

Variables, values, expressions, and printed output

- 01 Explain lists and dictionaries in plain language.
- 02 Use conditions in a small worked example.
- 03 Complete the guided lab and verify functions.

CORE CONCEPT

The vocabulary of this topic

Represent collections of values and control which instructions execute.

lists and dictionaries: A list stores ordered values by position; a dictionary stores values under meaningful keys.

conditions: A condition evaluates to true or false and controls which branch of code executes.

loops: A loop repeats a block for each item or while a stated condition remains true.

functions: Functions package a named transformation behind explicit inputs, outputs, and side-effect boundaries.

FOUR CONNECTED IDEAS

How the concept works

01



**lists and
dictionaries**

02

conditions

03

loops

04

functions

Collections hold related values; control flow selects or repeats work; functions package reusable transformations.

CORE CONCEPT

Read every part of the mechanism

Collections hold related values; control flow selects or repeats work; functions package reusable transformations.

list: An ordered mutable collection.

dictionary: Named keys mapped to values.

function: Reusable code with inputs and an output.

WORKED EXAMPLE

Worked example by hand

Filter three student scores, keep values at least 50, and compute the pass count.

01

Known values

02

Apply the mechanism

03

Interpret the result

RUN IT AND INSPECT THE RESULT

Map the concept to Python

```
scores = [42, 68, 91]  
passed = [score for score in scores if score >= 50]  
print(len(passed), passed)
```

OUTPUT

2 [68, 91]

Each line corresponds to a concept already explained.

QUALITY GATE

Build the complete mental model



Choose a list for ordered items, a set for uniqueness, and a dictionary for named fields.



``if`` selects; ``for`` repeats over items; ``while`` repeats while a condition stays true.



A function contract states accepted inputs, returned output, and possible failures.



Decompose a program into small functions that each perform one understandable transformation.

FOUR CONNECTED IDEAS

Notebook readiness map

01

**lists and
dictionaries**

02

conditions

03

loops

04

functions

Use the concepts from this lesson to read and complete
<https://github.com/curiously/Al-Bootcamp/blob/master/01.python-essentials-for-ai.ipynb>.

RISK REVIEW

Common beginner mistakes

01 **Confusing a list position with a dictionary key.**

02 **Printing inside every function instead of returning a reusable value.**

GUIDED LAB

Build a function that summarizes names and scores stored in dictionaries.

01

Predict

02

Run

03

Inspect

04

Explain

SESSION SYNTHESIS

Keep the system, not isolated definitions.

01 Explain lists and dictionaries in plain language.

02 Use conditions in a small worked example.

03 Complete the guided lab and verify functions.

NEXT EVIDENCE

A Python program using a list, dictionaries, a condition, a loop, and a function with a return value.